

Additional Content on PWM

Dr. Md. Rukonuzzaman

Associate Professor

Department of Electrical and Electronic Engineering
American International University- Bangladesh (AIUB)

PWM Type	Waveform Generation Mode	WGMn2	WGMn1	WGMn0	TOP	Channel A	Channel B	Frequency	Duty Ratio		Timer Counting
Fast PWM	3	0	1	1	255 (Timer 0/2) 65535 (Timer 1)	OCnA	OCnB	$\frac{f_{clk_IO}}{N(TOP + 1)}$	Non-inverting mode	$\frac{OCRnx}{(TOP + 1)}$	Up
									Inverting mode	$1 - \frac{OCRnx}{(TOP + 1)}$	
	7	1	1	1	OCRnA		OCnB		Non-inverting mode	$\frac{OCRnB}{(TOP + 1)}$	Up
									Inverting mode	$1 - \frac{OCRnB}{(TOP + 1)}$	
Phase Correct PWM	1	0	0	1	255 (Timer 0/2) 65535 (Timer 1)	OCnA	OCnB	$\frac{f_{clk_IO}}{2 \times N \times TOP}$	Non-inverting mode	$\frac{OCRnx}{TOP}$	Up & down
									Inverting mode	$1 - \frac{OCRnx}{TOP}$	
	5	1	0	1	OCRnA		OCnB		Non-inverting mode	$\frac{OCRnB}{TOP}$	Up & down
									Inverting mode	$1 - \frac{OCRnB}{TOP}$	

OCRnA: Output Compare Register A for Timer n(0/1/2);

OCnA: Output in Channel A for Timer n (0/1/2);

OCnB: Output in Channel B for Timer n (0/1/2);

OCRnx: Output Compare Register x (A/B) for Timer n (0/1/2);

N: Prescaler (1/8/64/256/1024); **f_{clk_IO} :** System Clock Frequency, Hz

¹¹ In inverting mode, d = OCR0A/(OCR0A+1) = 1; In Non-inverting mode, d = 1-OCR0A/(OCR0A+1) = 0

¹² In inverting mode, d = OCR0A/OCR0A = 1; In Non-inverting mode, d = 1-OCR0A/OCR0A = 0

Timer	Output Channel	Pin number in Arduino Uno	Pin Number and Port in Atmega328P
Timer0	OC0A	Arduino Pin 6	Chip Pin 12, PD6
	OC0B	Arduino Pin 5	Chip Pin 11, PD5
Timer1	OC1A	Arduino Pin 9	Chip Pin 15, PB1
	OC1B	Arduino Pin 10	Chip Pin 16, PB2
Timer2	OC2A	Arduino Pin 11	Chip Pin 17, PB3
	OC2B	Arduino Pin 3	Chip Pin 5, PD3

